

Each question is worth 5 points for a total of 30 points. The Extra Credit problem can get you up to an additional 5 points.

Problem 1

Advertisements for a certain small car claim that it floats in water. (a) If the car's mass is 900 kg and its interior volume is 3.0 m^3 , what fraction of the car is immersed when it floats? You can ignore the volume of steel and other materials. (b) Water gradually leaks in and displaces the air in the car. What fraction of the interior volume is filled with water when the car sinks?

Problem 2

An object is undergoing SHM with period 0.900 s and amplitude 0.320 m. At $t = 0$ the object is at $x = 0.320$ m and is instantaneously at rest. Calculate the time it takes the object to go (a) from $x = 0.320$ m to $x = 0.160$ m and (b) from $x = 0.160$ m to $x = 0$ m.

Problem 3

At point A , 3.0 m from a small source of sound that is emitting uniformly in all directions, the sound intensity level is 53 dB. (a) What is the intensity of the sound at A ? (b) How far from the source must you go so that the intensity is one-fourth of what it was at A ? (c) How far must you go so that the *sound* intensity level is one-fourth of what it was at A ? (d) Does intensity obey the inverse-square law? What about *sound* intensity level?

Problem 4

A Carnot heat engine has a thermal efficiency of 0.650, and the temperature of its hot reservoir is 900 K. If 3000 J of heat is rejected to the cold reservoir in one cycle, what is the work output of the engine during one cycle?

Problem 5

An aluminum tea kettle with mass 1.75 kg and containing 2.10 kg of water is placed on a stove. If no heat is lost to the surroundings, how much heat must be added to raise the temperature from 15.0°C to 90.0°C? Assume the specific heat of aluminum is 910 and the specific heat of water is 4190.

Problem 6

How much heat does it take to convert 1 kg of ice at -10.0°C to steam? Assume water has a heat of fusion of 3.34×10^5 , heat of vaporization of 2.26×10^6 , specific heat of 4190, and that ice has a specific heat of 2100.

Extra Credit:

Horseshoe bats emit sounds from their nostrils and then listen to the frequency of the sound reflected from their prey to determine the prey's speed. A Horseshoe bat flying at speed v_{bat} emits sound of frequency f_{bat} ; the sound it hears reflected from an insect flying toward it has a higher frequency f_{refl} . (a) Show that the speed of the insect is

$$v_{insect} = v \cdot \frac{f_{refl} \cdot (v - v_{bat}) - f_{bat} \cdot (v + v_{bat})}{f_{refl} \cdot (v - v_{bat}) + f_{bat} \cdot (v + v_{bat})}$$

where v is the speed of sound. (b) If $f_{bat} = 80.7$ kHz, $f_{refl} = 83.5$ kHz, and $v_{bat} = 3.9$ m/s, calculate the speed of the insect.